

AML Project Group 5 Technical Report

Predicting Unprofitable Health Insurance Policies

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and 4 team members

1. Executive Summary

This project analyzes a Spanish health insurance portfolio from 2017 to 2019. The objective is to identify policies that are likely to become unprofitable, where a policy is defined as unprofitable when annual claim cost exceeds annual premium.

With reference to our initial proposal, our final modeling interpretation shifted from fully automated classification to an early-warning risk prioritization approach for existing policies. Direct 0/1 prediction remains challenging across the models we tested, possibly because the dataset does not contain richer clinical details such as diagnoses, treatment types, or medical severity indicators. However, the models still show useful ranking ability, which can support the prioritization of policies for pricing or underwriting review.

The final workflow is designed to avoid using future test information too early. The 2017 records are used only to create prior-year lag features for the 2018 records, not as supervised training examples. After the lag features are created, only rows with valid prior-year history are used for modeling. The 2018 data is then split into a development-training set and a validation set: models are fitted on the development split, while model selection and the review threshold are based only on the 2018 validation results. The 2019 data is kept aside until the model and threshold rule are fixed and is used only for the final check on unseen data.

In the 2018 validation comparison, XGBoost and LightGBM performed almost the same. The differences across F1, PR-AUC, and ROC-AUC were very small, appearing only in the third or fourth decimal place. At each model's chosen threshold, XGBoost was slightly ahead on F1 (0.4352 vs 0.4347) and PR-AUC (0.3936 vs 0.3925), while LightGBM was slightly ahead on ROC-AUC (0.7408 vs 0.7406), which measures overall ranking quality without depending on a specific threshold. Since our goal is not direct binary classification but risk prioritization, we selected LightGBM as the operating model for the decile/lift analysis and fixed-threshold checks. As a secondary consideration, LightGBM's significantly lighter computational footprint and faster training speeds ensure maximum efficiency when deployed.

Using the 2018 validation decile/lift table, the top 30% boundary was selected as a workable policy review cut-off. The LightGBM score at that boundary, 0.428926, was fixed as the deployment threshold. When this same threshold was applied once to the 2019 test set, it flagged 14,621 of 52,455 policies, or 27.9%, captured 4,415 of 8,124 unprofitable policies, and achieved approximately 30.2% precision, 54.3% recall, and 1.95x lift.

2. Business Problem and Data Overview

The insurer wants to identify policies whose claim costs are likely to exceed premiums. These policies can create underwriting losses if they are renewed, priced, or monitored without additional review. However, detailed manual review across an entire policy portfolio is often resource-intensive and may not be practical on a frequent basis. Therefore, the model is positioned as an early-warning risk prioritization tool rather than an automatic decision system. By ranking existing policies according to predicted unprofitability risk and being selective on which portion to review, the model helps insurers focus pricing review, underwriting intervention, product targeting, and portfolio monitoring on the cases most likely to require attention. The project objective is:

Build a machine learning pipeline that predicts the probability that a policy is unprofitable and ranks policies by risk, so high-risk policies can be automatically prioritized and flagged to increase policy-review efficiency.

Component	Definition
Input X	Policyholder, policy, geographic, socioeconomic, climate, clustering, and lagged historical features
Target Y	1 if annual claim cost exceeds annual premium, otherwise 0
Problem type	Unsupervised clustering followed by supervised binary classification and ranking
Business use	Early-warning review list for pricing and underwriting support

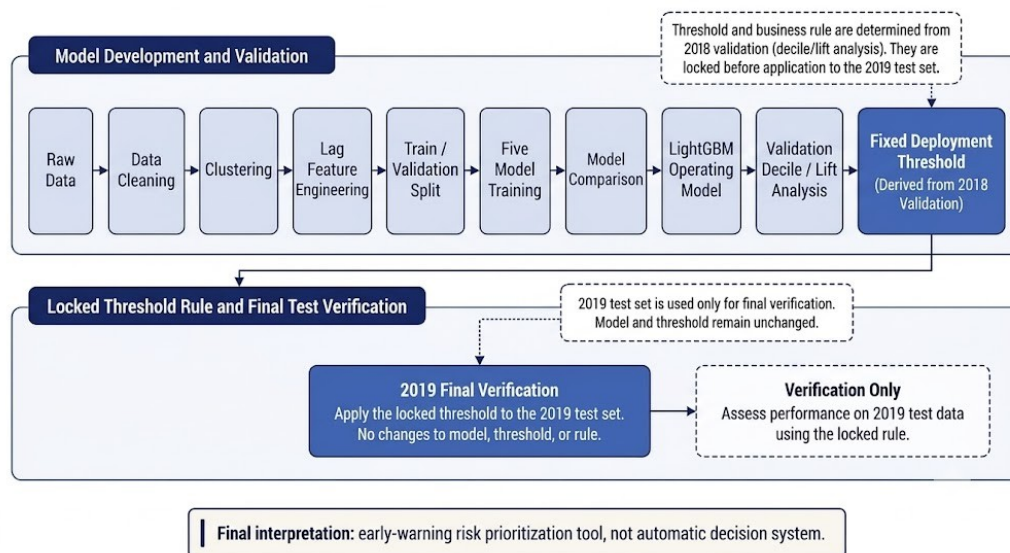
The main dataset is Dataset of health insurance portfolio.xlsx. It contains 228,711 insured-person-year records and 42 raw variables across 2017, 2018, and 2019. The dataset includes policy attributes, customer demographics, premium, claim cost, number of medical services, geographic variables, socioeconomic indicators, and a climatological cluster variable. The modeling pipeline further adds engineered clustering labels and lagged historical features.

The target variable is imbalanced, with unprofitable policies forming a minority of the portfolio. In the final 2019 test set, 8,124 out of 52,455 policies are unprofitable, giving a base unprofitable rate of 15.5%. The 2018 validation set has a slightly higher base rate of 17.8%. This difference matters because precision is affected by how common unprofitable policies are in each sample, so the 2019 test precision should be interpreted considering the lower underlying unprofitable rate.

Data Split / Scope	Value
Raw records	228,711
Raw variables	42
2018 validation policies	13,114
2018 validation unprofitable policies	2,334 (17.8%)
2019 final test policies	52,455
2019 unprofitable policies	8,124 (15.5%)

3. Final Pipeline Overview

The final pipeline follows the same sequence as the technical workflow. It first prepares leakage-safe data, then adds segmentation and prior-year lag features, compares candidate models using the 2018 validation set, and finally fixes a deployment threshold before evaluating the 2019 test set.



The key methodological principle is that the 2019 set is not used for model selection, threshold tuning, or rule design. It is used only once, after the operating model and deployment threshold have been fixed, to provide final validation on unseen data.

4. Data Cleaning and Leakage Control

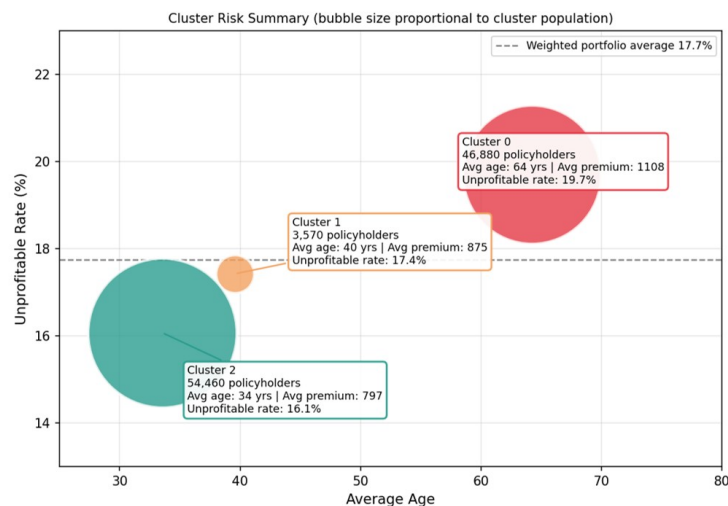
Data cleaning focused on creating a valid prediction setting. The main filters and controls were:

Step	Decision	Reason
Product scope	Keep Standard and Premium products	These are the main comparable health insurance product tiers; Dental and International products are smaller and structurally different.
Time consistency	Use 2017 only for prior-year lag construction, develop the model on 2018 set, and reserve 2019 as the future test year	Mimics a real deployment setting and prevents look-ahead bias.
Target creation	Define unprofitable: cost_claims_year > premium	Matches the business definition of claim-cost risk.
Leakage removal	Exclude current-year cost_claims_year	It directly defines the target.
Leakage removal	Exclude current-year n_medical_services	It is only known after medical usage occurs during the policy year.
Leakage removal	Exclude exposure_time	Current-year exposure may not be known at renewal/start-year scoring and may indirectly reflect lapse behavior.
Identifier handling	Use IDs only for lag construction	IDs are needed to link prior-year records but should not be used as predictive model features.

Lagged historical claim features are used because they come from information already known from the previous year. For example, a 2019 prediction may use 2018 claim history, but it must not use 2019 claim cost, medical service count, or exposure information.

5. Clustering Methodology

Clustering was used for customer profiling and feature engineering, not as a final prediction model. We tested K-Means, Gaussian Mixture Models, and DBSCAN, and selected K-Means with $k = 3$ because it achieved the strongest silhouette score and produced interpretable customer segments. The clustering results show meaningful differences across policyholder groups: Cluster 0 represents an older, higher-premium segment with the highest unprofitable rate; Cluster 1 represents a smaller mid-age group with risk close to the portfolio average; and Cluster 2 represents the largest, younger, and lowest-risk segment. The resulting cluster_id was then used as an additional feature in the supervised models, rather than as a standalone decision rule.



6. Feature Engineering

Feature engineering includes original policyholder variables, encoded categorical variables, clustering labels, and lagged historical features. The lag features are especially important because prior-year claim behavior is a legitimate and

predictive signal. As such, we used 2017 to provide history for 2018 and used 2018 for our model development. As mentioned, data from 2019 is disregarded during training, and would only be used for final testing.



Lag Feature	Meaning
lag1_premium	Previous year premium
lag1_cost_claims_year	Previous year claim cost
lag1_claim_ratio	Previous year claim cost divided by previous year premium
lag1_unprofitable	Whether the policy was unprofitable in the previous year
lag1_n_medical_services	Previous year medical service count
lag1_cluster_id	Previous year cluster label
premium_change_abs	Absolute premium change from previous year
premium_change_pct	Percentage premium change from previous year
lag1_claim_to_premium_gap	Previous year claim cost minus previous year premium

7. Modeling Approach

The project evaluated five supervised learning models. Each model notebook tunes hyperparameters and thresholds using a 2018 development/validation split, then exports validation metrics for model comparison. The 2019 test set is not used to select models, tune thresholds, or choose the deployment threshold.

Model	Pros	Cons
Logistic Regression	- Fast to train, even on large datasets - Coefficients tell you the direction and magnitude of each feature effect	- Struggles with complex, non-linear patterns - Sensitive to correlated features (multicollinearity)
Decision Tree	- Highly interpretable; you can visualize the exact decision rules - Handles non-linear relationships naturally	- Overfits easily with suboptimal hyperparameters - Unstable; small changes in data produce very different trees
Random Forest	- Reduces overfitting by averaging many trees - Handles non-linear relationships well	- Slower to train and predict than a single tree or logistic regression - Less interpretable than a single decision tree
XGBoost	- Often achieves the best predictive performance on tabular data - Highly tunable for squeezing out performance	- Many hyperparameters to tune; requires more effort to set up properly - Less interpretable than a single decision tree

The validation design prevents test-set leakage. Model selection is based on 2018 validation outputs. Only after the operating model and deployment threshold are fixed is the 2019 test set used for final verification.

8. Validation Model Results

The model comparison shows that the tree-based models perform similarly. This suggests a realistic performance ceiling: the available variables contain useful ranking signals, but they do not support accurate direct classification.

Model	F1	Precision	Recall	ROC-AUC	PR-AUC	Threshold
XGBoost	0.4352	0.3476	0.5818	0.7406	0.3936	0.56
LightGBM	0.4347	0.3468	0.5823	0.7408	0.3925	0.43
Random Forest	0.4323	0.3296	0.6281	0.7363	0.3769	0.51
Decision Tree	0.4238	0.3262	0.6045	0.7263	0.3653	0.56
Logistic Regression	0.4047	0.3326	0.5167	0.7150	0.3579	0.52

XGBoost held a marginal lead in the F1-score, showing highly precise classification at a static threshold, as well as PR-AUC. However, LightGBM achieved a marginally superior ROC-AUC. Because our primary business objective is early-warning risk prioritization and relative risk ranking rather than direct, hard binary classification, a model's ability to discriminate and rank order risk across all thresholds is paramount.

Consequently, LightGBM was selected as the final operating model for decile/lift analysis and deployment-threshold verification. Its higher ROC-AUC matches our prioritization strategy, and its significantly lighter computational footprint and faster training speeds ensure high operational efficiency when deployed.

9. Evaluation Framework

F1-score serves as a technical benchmark for direct classification but is insufficient for the business decision because false negatives and false positives carry asymmetric costs: a missed unprofitable policy incurs recurring underwriting loss at every renewal, whereas a false positive costs only a one-off review. This asymmetry, combined with the 15–18% class imbalance, rules out accuracy and makes F1 alone inadequate.

Therefore, the final evaluation emphasizes ranking quality and operating-point performance rather than a single threshold-bound score. Ranking metrics (ROC-AUC, PR-AUC) are threshold-independent and measure whether the model orders policies by risk, which is basically the property that makes any downstream cut-off viable. Decile lift and cumulative gain translate that ranking into business terms: how concentrated losses are at the top, and how many are captured at a given review capacity. Precision reports the flagged-list quality at the chosen operating point.

Metric	Role
F1-score	Technical benchmark for direct classification
ROC-AUC	Overall ranking ability across thresholds
PR-AUC	Ranking performance under class imbalance
Decile lift	Whether high-risk ranked groups contain more unprofitable policies than average
Cumulative gain / recall	How many unprofitable policies are captured by the flagged group
Precision	How clean the flagged review list is

10. LightGBM Validation Decile/Lift Interpretation

The decile/lift table is computed on the 2018 validation set before the 2019 test set is touched, making the deployment threshold data driven. Each row represents a review capacity that the company can commit to, and shows the expected volume, precision, lift, and share of unprofitable policies captured.

Review Capacity	Decile	Policies Reviewed	Unprofitable Found	Precision	Lift	% Unprofitable captured
Top 10%	1	1,312	615	46.9%	2.63x	26.3%
Top 20%	2	2,623	1,017	38.8%	2.18x	43.6%
Top 30%	3	3,935	1,362	34.6%	1.94x	58.4%
Top 40%	4	5,246	1,606	30.6%	1.72x	68.8%
Top 50%	5	6,557	1,828	27.9%	1.57x	78.3%
Top 60%	6	7,869	1,987	25.3%	1.42x	85.1%
Top 70%	7	9,180	2,132	23.2%	1.30x	91.3%
Top 80%	8	10,492	2,229	21.2%	1.19x	95.5%
Top 90%	9	11,803	2,291	19.4%	1.09x	98.2%
Top 100%	10	13,114	2,334	17.8%	1.00x	100.0%

The table here is the key deliverable: the business can decide on any review budget and know the expected return. Because lift decreases monotonically, coverage is most efficient at the top and capacity for the top 30% already captures

58.4% of unprofitable policies, and 60% captures 85.1%, with diminishing returns as lift approaches the no-skill baseline (1.0x).

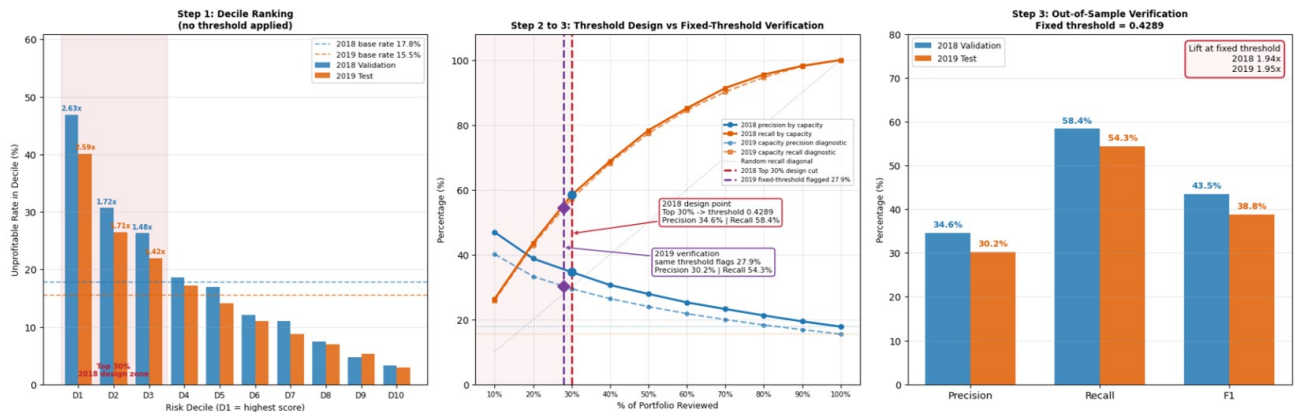
We adopt the top 30% boundary as the operating point indicating that the inflection where lift falls under 2x and per-decile gains begin to shrink. It reviews 3,935 policies, captures 1,362 unprofitable ones at 34.6% precision ($\approx 2\times$ the 17.8% base rate) and 1.94x lift. The LightGBM score at this boundary, **0.428926**, is fixed as the deployment threshold.

11. Fixed Threshold Verification on 2019 Test Set

Rather than optimizing a threshold on the test set, we fix the operating point on the 2018 validation set and verify it in 2019, preventing test-set leakage. Validation policies are ranked by LightGBM score and partitioned into risk deciles; the decile-lift profile shows loss concentration monotonically decreasing from decile 1, remaining above 1.3x through the top three deciles (top 30%) before decaying toward the no-skill baseline (1.0x). We therefore adopt the top 30% validation boundary as the deployment cut-off. The corresponding score, which is recorded from the LightGBM probability at the 30th-percentile rank of the 2018 validation distribution is $\tau = 0.428926$. The deployment rule is flag policy i for review if $\hat{P}(\text{unprofitable})_i \geq \tau$.

We apply the frozen τ to 2019 without recalibration. Critically, the flagged volume is **not** forced to 30%; it is determined endogenously by τ and the 2019 score distribution. This simulates true deployment, where a fixed rule meets a shifting population, and tests whether the model's ranking generalizes out-of-time.

Metric	2018 Validation Threshold Design	2019 Test Fixed-Threshold Verification
Deployment threshold	0.428926	0.428926
Total policies	13,114	52,455
Base unprofitable rate	17.8%	15.5%
Flagged policies	3,935	14,621
Flagged policy rate	30.0%	27.9%
Unprofitable policies captured	1,362	4,415
Precision / flagged-list unprofitable rate	34.6%	30.2%
Recall / captured unprofitable pct	58.4%	54.3%
Lift	1.94x	1.95x
Ranking ROC-AUC	0.7408	0.7265
Ranking PR-AUC	0.3925	0.3315



Under the frozen threshold, 2019 flags 27.9% of policies which are close to the 30% design target despite a lower base rate (15.5% vs 17.8%) and no retuning. The flagged list captures 4,415 of 8,124 unprofitable policies at 30.2% precision and 54.3% recall, roughly 2x the base rate. Lift is stable (1.95x vs 1.94x) and ROC-AUC holds (0.727 vs 0.741), confirming the ranking generalizes out-of-time; the PR-AUC drop (0.393→0.332) reflects the lower positive prevalence, as expected for a prevalence-sensitive metric. The consistency of lift and precision across years validates the fixed rule as a repeatable, deployable early-warning screen rather than an artefact of the validation split.

12. Business Application: Early-Warning Risk Prioritization Workflow

To operationalize the trained LightGBM model, we propose a standardized, human-in-the-loop "co-pilot" workflow for the underwriting and pricing teams. This framework is specifically designed to assist, rather than automate, business decisions by maximizing efficiency in the policy review processes. The day-to-day workflow consists of the following five key steps:

1. **Score:** Run active, renewing policies through the trained LightGBM operating model.
2. **Flag:** Automatically isolate policies that return an unprofitability risk score above 0.428926.
3. **Rank:** In scenarios where review capacity is limited, sort the flagged policies from highest to lowest risk score to establish an objective triage queue.
4. **Act:** Instruct underwriters to apply localized strategic judgment to the prioritized queue, combining the model's risk score with churn propensity signals and customer segments to decide whether to reprice, monitor, adjust product targeting, or accept natural churn.
5. **Track:** Log subsequent claim outcomes over the active policy year to feed the historical training database for annual model and threshold recalibration.

As mentioned in the previous section, an analyst executing this workflow in 2019 would successfully narrow the manual review volume down from 52,455 policies to a highly targeted segment of 14,621 policies (27.9% of the portfolio). By auditing this prioritized 27.9% segment, the team would be able to identify and address around 54.3% of total portfolio losses. This operational setup is vastly more efficient and resource-effective than conducting blanket audits across the entire portfolio.

13. Limitations

While the LightGBM pipeline represents a major improvement in the efficiency of portfolio risk management process, its successful governance requires a clear understanding of its inherent technical and operational boundaries:

- **Absence of Clinical Detail:** The baseline dataset contains no granular clinical indicators, such as diagnostic codes, chronic disease markers, or medical severity metrics. This limits absolute classification accuracy, reinforcing why the model must be used for relative risk ranking rather than automated binary decisions.
- **Calibration Limitations:** Due to the class-weighting adjustments applied during model training to manage dataset imbalance, the output scores do not reflect absolute, real-world probability. They must be interpreted strictly as ordinal, relative risk rankings.
- **Fluctuating Review Volumes:** Utilizing a fixed deployment threshold of 0.428926 does not guarantee a static annual review volume. If the underlying score distribution of the portfolio shifts, the number of flagged policies will change.
- **Unquantified Cost-Benefit Metrics:** A fully optimized mathematical cost-benefit model is currently out of reach because the individual cost of a manual underwriting review, the precise financial benefit of an intervention, and the exact losses avoided per true positive have not yet been financially quantified.
- **Performance Drift:** Shifting customer claim behaviors, evolving pricing rules, and macro portfolio composition changes over time will degrade predictive power. The model must undergo annual monitoring and recalibration.

14. Conclusion

Deploying this machine learning pipeline successfully shifts our risk-management strategy from reactive damage-control to a proactive and highly efficient approach. This can deliver immediate operational value across four distinct dimensions:

- **Bypassing the Stable Core 70%:** The algorithm automatically filters out lower-risk, healthy policy accounts. This allows underwriters to confidently skip 70% of the portfolio, eliminating unnecessary administrative audits.
- **Elimination of Manual Triage:** Rather than wasting valuable hours manually sorting through spreadsheets and claim databases, underwriting teams can access the pre-sorted, prioritized list via a Portfolio Action Dashboard.
- **Doubling Review Productivity:** Verified against unseen 2019 test data, our fixed deployment threshold achieved a strong 1.95x model lift. This means every hour an underwriter spends analyzing the model's flagged policies is nearly twice as productive, capturing 54.3% of all unprofitable policies while evaluating less than 28% of the total portfolio.
- **Proactive Interception:** The framework intercepts chronically unprofitable policies *at renewal*, before claims are filed and paid out. This empowers pricing teams to implement strategic adjustments or accept natural churn proactively, stopping silent financial leakage before it compounds.

Portfolio Action Dashboard prototype:

Underwriters can utilize this dashboard to get a clean, user-friendly list of risky policies.

The interface displays immediate risk ranking scores, visualizes key financial indicators like premiums versus claims, and automatically serves pre-calculated business action recommendations such as Reprice, Accept Churn, or Retain based on the policy profile clustering done earlier in the process

